Wildling: Product Requirements Document V1

Submitted to the project lead · Product, interaction design and implementation plan · Responds to Brief V1 Date 2026-10-09 · Status: submitted for review

V1 and V2 are requirement stages, not code versions. The code explorations are Prototype A (`prototypes/a-autonomous-life/`) and Prototype B (`prototypes/b-care-tools/`). Features, parameters and acceptance targets become the development baseline after review.

Source tags

Tag	Meaning	Who decides
[Client]	A goal, boundary or requirement the project lead set in the brief	Project lead
[Proposal]	A product, interaction, parameter or technical decision I propose	I propose; aligned in review
[To confirm]	A trade-off affecting scope, schedule or experience that both sides must confirm before it enters the baseline	Both

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1 My reading of the brief

[Client] An interactive piece on "how care builds trust" that is understandable, alive, relational and presentable on its own.

[Proposal] Visitors need to feel two things at once: "its response is about what I did", and "it does not revolve around me". The product therefore needs **understandable input, legible needs and a degree of autonomy**.

The biggest design risk runs in two opposite directions:

- the character looks rich, but visitors do not know what they can do;
- the controls are clear, but the character feels like an animation answering buttons.

Explicit tools lower the first-time learning cost; rest, fullness and approach boundaries keep the character's own rhythm. I **do not recommend** delivering the full ecology, generations and complex skeleton in round one. They add atmosphere but do not prove the core relationship works. Success for round one: a visitor completes one effective act of care, reads one autonomous choice, and understands one growth change.

2 Product claim and design hypotheses

[Client] The product claim is "how care builds trust".

[Proposal] Three kinds of care map to fullness, fun and trust; rest maps to energy; growth marks time spent together. These are hypotheses to validate, not conclusions:

ID	Hypothesis	How it is tested
H1	Explicit tool entry points are easier for first-time visitors than relying only on natural gestures	Feed, Play and Pet entry points; first-visit usability test (A1-10)
H2	Visitors read the relationship through distance, ear and tail posture and sound; meters only support	Build state expression before fur, lighting and props
H3	A refusal that has a reason and recovers quickly does not break the experience, and may make the creature feel more alive	First-visit observation
H4	Three environments help visitors read growth; a full life cycle is not needed for a satisfying close	Show a care recap at adulthood; elders, breeding and genetics later

3 From brief to product

The B1 IDs are mine, added for traceability.

ID	[Client] Brief goal	[Proposal] Product response
B1-01	Care builds trust	A clear causal chain from tool to need to relationship feedback
B1-02	First-time visitors understand	Three visible tools, mouse fallback, one-line instant hints
B1-03	The creature acts on its own	Autonomous exploring, expressing needs and resting; allowed to refuse ineffective care
B1-04	Gentle and fast actions get different responses	Position, distance and speed rules with smoothing and valid-input checks
B1-05	Growth and three environments are visible	Kitten, juvenile and adult chapters with transitions and a recap
B1-06	Visitors understand the reasons for behaviour	Body language plus short bubbles; developer view shows state and needs
B1-07	Presentable on its own	Browser build, real camera test, mute and restart
B1-08	Fits creative coding	JavaScript runs the interaction; Processing Java validates visual and motion rules

4 Users and setting

- **[Client]** First-time members of the public, one operator per round with others watching, on a computer with a webcam, mouse and speakers.
- **[Proposal]** Target display range 1280×720 to 1920×1080.
- **[Proposal]** About 3–5 minutes per round as the prototype's rhythm target. Visitors can end before growth completes; speed is not a success measure.

- **[Proposal]** Round one is a demonstrable prototype. Unattended all-day operation, multi-visitor control and arbitrary lighting are later-stage goals.
- **[To confirm]** Actual devices, display size, and left/right hand and sitting/standing test conditions.

5 Scope and route choice

5.1 In scope for this round [Proposal]

Start and input choice; camera hand position and pinch confirm; mouse fallback; Feed, Play and Pet; fullness, fun, energy and trust; autonomous exploring and rest; avoidance of fast close approaches; three-chapter growth; state hints, sound and mute; care recap and restart; source code, run notes and demo material.

Processing Java delivers a runnable **visual and motion study** this round, replaying the same parameters and anonymous state. The interactive prototype is browser-first. Wiring live perception into Processing and the exhibition main screen is the next stage.

5.2 Deferred [Proposal]

Autonomous hunting, full prey ecology, eight-joint spine and two-bone IK, genetics, elders and rebirth, persistent profiles, a newly trained gesture model, open dialogue and cloud generation. Prototype A's systems are research material for these later stages.

5.3 Three routes compared [Proposal]

Route	Strength	Trade-off
Natural gestures + autonomous ecology (Prototype A)	Minimal UI, strong sense of observation	Higher barrier for first-timers; not chosen for round one
Three tools + visible needs (Prototype B)	Easy to understand; care causality is testable	**Chosen as the main route**, with autonomous rest to avoid feeling mechanical
Free conversation	Expresses personality easily	Adds latency and content scope without solving action understanding; deferred

The cost of the chosen route is that visitors switch tools. My answer: **the tool states what the visitor is about to do; the character's state decides whether it accepts.** No button is a guaranteed command.

6 Experience flow and interface [Proposal]

1. **Entry:** a short introduction and two choices, camera or mouse. Camera permission is requested only after choosing the camera; the mouse path loads no hand model. First entry shows only the current tool and one line of guidance.
2. **During:** observe its needs, choose a kind of care, adjust to body language and bubbles. With low energy it rests; when full it refuses food; with low trust it keeps more distance. With basic care met, body and environment change.
3. **Close:** at adulthood or on request, show effective feeding, play and contact counts and a line about the relationship. The line comes from fixed rules, **with no generative text and no moral judgement of the visitor**.

Layout: the character and world in the centre; three tools at the bottom; stage and needs top-left; input status and mute top-right; short bubbles near the character. Meters show relative change; exact values are for the developer view.

[To confirm] Public copy in English ("Hold still to pet", "I'm full", "I need a nap"), with a second language in a separate copy file. Whether a formal language switch ships is confirmed at kick-off.

7 Input and interaction spec [Proposal]

7.1 One input object

Camera and mouse produce one object: `source, x, y, present, speed, confirm, timestamp`. Character logic reads only this object, so mouse and camera never have two sets of care rules.

- Positions use content coordinates 0–1; speed uses **S/s** (short screen side per second); mirroring happens once, at input.
- Hand keypoints not updated for 300 ms count as lost and stop counting contact.
- Speed resets when a hand first appears or the input switches, so jumping in from an invalid position is never read as a scare. **A tracking jump is a system issue and never costs trust.**

7.2 Rules and starting parameters

Item	Start value	Meaning and validation
Pinch close ratio	thumb–index gap / palm size < 0.25	One confirm; validate with real hands
Pinch release ratio	> 0.45	Hysteresis prevents repeats
Confirm cooldown	400 ms	A held pinch confirms once
Gentle movement reference	< 0.40 S/s	Combined with tool, distance and dwell for contact
Fast close reference	> 1.10 S/s and < 0.18 S from the cat	Only valid trajectories count toward alarm
Contact dwell	600 ms	Passing over the body does not start petting

These are starting values to tune, not measured optima. Hand shape is used only for pinch and keypoint validity this round; "open hand means petting" is guidance only. Understanding hand shape is the research question for the next stage. Toolbar hits take priority over the scene.

8 Feature spec [Proposal]

ID	Feature	Rule	Boundaries and anti-farming
F1-01	Feed	Up to 3 fish on the ground; the cat seeks the nearest when fullness < 0.85, eats for 1.2 s; on completion fullness +0.25, trust +0.025	At fullness ≥ 0.85 it refuses without affecting trust; the recap counts fish *eaten*, not dropped; uneaten fish fade after 30 s

ID	Feature	Rule	Boundaries and anti-farming
F1-02	Play	With energy > 0.30 it chases; reaching the toy is one pounce: fun +0.12, energy −0.04, trust +0.01	At least 1.5 s between pounces, never per frame; moving the toy away costs nothing
F1-03	Pet	Slow dwell of 600 ms on the body requests contact; accepted when not asleep, not startled and fullness ≥ 0.20; trust +0.02 per second	At most 6 s per contact, then a pause and reposition; a lost hand stops the timer; a refusal costs no trust
F1-04	Observe and autonomy	Without care it explores, pauses, signals hunger or grooms; at energy < 0.15 it rests until 0.65	No death from neglect; no guilt-inducing copy
F1-05	Fast approach	Low trust + a valid close trajectory + speed condition → a brief retreat; the first time only shows alertness	Only 2 s of sustained valid scaring may reduce trust a little, at most 0.05 per episode; lost hands, jumps, input switches and the first recovered frame never count

9 Needs model and behaviour selection [Proposal]

- Starting values: fullness 0.55, fun 0.50, energy 0.80, trust 0.25, all clamped to 0–1.
- Decay: fullness −0.004/s, fun −0.003/s, energy −0.003/s; resting +0.035/s. All in real seconds, independent of frame rate.
- **Priority state machine:** pause and end first, then valid scare, needed rest, finishing a meal, urgent hunger, valid contact, seeking food, play, low-need hints and autonomous exploring. States hold a minimum duration so bubbles and posture never flicker within a second.

> **Why a state machine, not Prototype A's utility AI, for round one?** With few behaviours, causality and readability come first. Utility AI suits later stages where many needs compete. Round one does not need a decision system visitors cannot read just to look "more AI". Machine learning is used for HandPose keypoints; character decisions are explainable rules. (ADR-002)

10 Growth and recap [Proposal]

- Wellbeing = mean(fullness, fun, energy). Growth advances when wellbeing > 0.50, at 1/80 per second × (1 + 0.25 × trust). 0–1 kitten in the jungle, 1–2 juvenile in the snowfields, 2–3 adult in the desert; transitions take about 1.5 s.
- **Known risk:** an average can hide one need being very low. Replays and tests check for this; if "one need is very low but it still grows" appears, the next round adds a per-need minimum (adopted in PRD V2 F2-13).
- The recap shows effective feeds, successful pounces, accepted contact time and one relationship line. With no successful care, a neutral close: "We shared a quiet moment."
- Speed-up exists only in an explicit Demo mode with an on-screen label; it is disabled in real sessions and testing.

11 AI and technical approach [Proposal]

- JavaScript and p5.js run the main loop; ml5.js HandPose provides keypoints (loaded only when the camera is chosen); Web Audio synthesises all sound; camera failure switches to the mouse immediately.
- Processing Java is a separate motion and visual study: PVector movement, turning, tail, gait and rest pose, replaying anonymous synthetic state JSON. **It runs no second set of needs or growth.**
- Modules: InputAdapter, CreatureState, BehaviourController, WorldView, SoundAdapter, SessionController and UI, with parameters and copy kept separately.
- At least one event replay is kept: input, behaviour changes, effective care and state changes. It is the baseline for moving to an authoritative core next stage.

12 Design expression and accessibility [Proposal]

- Each of explore, eat, chase, contact, retreat and rest has defined motion, posture, text and sound: contact = closed eyes or relaxed tail + purr; retreat = backing away + ears + a short hint; rest = curling up + slower breathing. **All key states stay readable with sound off.**
- Tools have selected states and text labels, not icons alone; danger and success never rely on red and green.
- **[To confirm]** The mouse is the full alternative input this round. Keyboard control, a no-pinch path and complete focus navigation are next-round accessibility work; if required in round one, they are added to scope with test time.

13 Testing and acceptance

[Proposal] These targets are my proposal for review.

ID	Scenario	Pass condition
A1-01	Full mouse session	Choose tools, care, rest, pass three chapters and restart
A1-02	Real camera	Position mapping, pinch and contact on the target device, live
A1-03	Camera declined	Choose the mouse directly; nothing locks up
A1-04	Repeated pinch and toolbar clicks	One pinch confirms once; tool clicks never drop food
A1-05	Full and tired	Refuses food and rests for matching reasons, then recovers
A1-06	Lost hand or jump during contact	Timer stops, no trust lost, no scare on the first recovered frame
A1-07	Watching only	Autonomous behaviour continues without clicks

ID	Scenario	Pass condition
A1-08	Mute and restart	Key feedback still readable; all relationship and care stats reset
A1-09	Processing motion replay	The same state reproduces the same pose, with no second relationship logic
A1-10	First usability test	At least 4 of 5 complete an effective act of care within 30 s and explain one piece of feedback
A1-11	Continuous demo	30 minutes and 5 restarts on the target device without blocking errors

Builds, devices, parameters, observed issues and before/after changes are logged.

14 Responsibilities

- **I own:** turning the brief into a product plan, interaction flow, interface and motion-feedback spec; building the browser prototype, the Processing study, sound, state logic and demo package; agreed testing, fixes and delivery documentation.
- **[Client] The project lead owns:** theme, main setting, devices, public copy, scope and phase acceptance, and support with venue conditions and test recruitment.
- Review aligns key scope; not every visual detail is locked up front. Parameters and local interactions are updated after prototype testing. New features, multi-visitor support or a full Processing main screen are scheduled separately.

15 Plan and deliverables [Proposal]

About 14–18 working days over 3–4 weeks, reusing existing prototype material, including small-scale testing and revision. Large new visual assets, model training and unattended venue delivery are outside this stage.

Checkpoint	Content
CP1 Input and state skeleton	Mouse loop, camera mapping and simple poses
CP2 Playable	Three kinds of care, refusal and recovery, three chapters and recap
CP3 Delivery	Processing replay, user testing, run notes and demo material

Deliverables: product and interaction notes, interface and motion-feedback spec, browser source and build, Processing study and replay data, parameter and copy configuration, test and issue log, run notes and a 60–90 s demo video. Reused code and third-party resources are credited with their scope of change.

16 Decisions for the project lead [To confirm]

ID	Decision	My recommendation	If not confirmed
Q1	A short session built on "three tools + autonomous rest", or keep full autonomous hunting and life cycle	The former; the latter as a later observation mode	Hunting and generations in round one need a separate estimate
Q2	Accept the browser as the main prototype and Processing as a visual and motion study	Accept	A Processing public screen in round one adds communication and dual-client integration
Q3	Accept ~3–5 minutes and an adulthood recap; English main UI with a second language; when devices, venue conditions and testers are available	Accept; English UI with copy in config	Affects copy and test schedule

Until Q1 and Q2 are confirmed, only reversible input validation and low-fidelity design proceed.

17 Triggers for the next iteration [Proposal]

If we observe	The next round should
Gestures often don't match intent	Add temporal features, quality gating and an evaluated gesture classifier
Visitors watch the trust bar instead of the character	Strengthen body language, reasons and refusals
Interaction feels like grinding tasks	Introduce event memory and bounded rhythm adaptation
The piece goes to a public exhibition	Full start, pause, exit, end and turnover rules; staff diagnostics; offline package; Processing main display; repeated long runs

These become inputs to Brief V2 and PRD V2.

18 References and document control

- Requirements: Brief V1. Technical material: Prototype A (`prototypes/a-autonomous-life/`) and Prototype B (`prototypes/b-care-tools/`).
- Technical references: p5.js, ml5.js, Processing reference, MDN getUserMedia. Dependency versions are pinned after verification.
- **Planned, implemented, verified and confirmed are four different states**, tracked separately in `../STATUS.md`.